

Recipe Chatbot — C-CS219 Functional Programming Project

A rule-based chatbot built in Scala that helps users discover recipes based on their dietary preferences.

Requirements

- Java JDK 8 or higher
- Scala 2.13
- sbt (Scala Build Tool)

Installation & Running

1. Clone the repository:

```
git clone https://github.com/ammarnabilhusein/recipe-chatbot.git
```

2. Navigate to the project folder:

```
cd recipe-chatbot
```

3. Run the chatbot:

```
sbt run
```

Usage

Type your message and press Enter. Example inputs:

- `hello` — greet the chatbot
- `i am vegan` — set dietary preference
- `suggest something` — get recipe recommendations
- `protein` — ask about nutrition topics
- `summarize` — get a conversation summary
- `what topics have we discussed` — see discussed topics
- `quit` — exit the chatbot

Architecture

File	Description
<code>DataStructures.scala</code>	Case classes: <code>Recipe</code> , <code>InteractionEntry</code> , <code>ConversationState</code> and the recipe knowledge base
<code>Chatbot.scala</code>	Core chatbot — handles user input, generates responses, pattern matching on intents
<code>Recommender.scala</code>	Recommendation engine — filters and ranks recipes based on user preferences
<code>Memory.scala</code>	Conversation memory — logs interactions, detects repeated queries, summarizes conversation
<code>Main.scala</code>	Entry point — runs the conversation loop recursively

Functional Programming Concepts Used

- Pattern matching for intent detection
- Immutable data structures (`List`, `Map`)
- Higher-order functions (`filter`, `map`, `foldLeft`, `groupBy`, `sortBy`)
- Pure functions with no side effects
- Recursive conversation loop instead of mutable while loop
- Option types for safe preference retrieval